

Distinction Self-Duality — Structural Reconstruction of Quantum Causality and 4D Gravity

Structural Baseline

Purpose

This document freezes the current structural theory before attempting microscopic calibration of Newton's constant G , the cosmological constant Λ , the microscopic record scale ℓ^* , or finite- ℓ correction coefficients. It distinguishes assumptions, derived results, known theorems invoked, the single dynamical postulate, continuum/regularity conditions, and open quantitative problems.

Status Key

[DEF] Definition.

[DER] Derived within the present framework.

[KNOWN] Established external mathematics invoked at the indicated step.

[POST] Fundamental postulate.

[COND] Continuum, regularity, or representation condition that must hold for the indicated theorem to apply.

[OPEN] Not yet derived.

1. Primitive Ontology: Distinction Before Spacetime

[DEF] Begin with a set X of possible physical states or contexts together with a primitive orthogonality/distinguishability relation $\perp$. No spacetime, coordinates, causal order, fields, or particles are assumed at this level.

[DEF] Irredundancy:

$$x\perp = y\perp \Rightarrow x = y.$$

Operationally, no two distinct states have exactly the same distinction profile.

[POST/STRUCTURAL PRINCIPLE] Distinction Self-Duality (DSD):

$$f(a) \perp b \Leftrightarrow a \perp f^\dagger(b).$$

The meaning is that every physically realizable distinction profile induced by a map f is internally representable by a unique adjoint image $f^\dagger$. Reality is self-contained: distinguishability need not be evaluated by an external observer.

[DER] Dagger laws follow:

$$I^\dagger = I,$$

$$(g \circ f)^\dagger = f^\dagger \circ g^\dagger,$$

$$(f^\dagger)^\dagger = f,$$

under irredundancy and profile representability.

[KNOWN/COND] Under the standard closure, point-closedness, irreducibility, orthomodularity, and representation hypotheses used in orthogonality-space reconstruction theorems, the DSD structure leads to projective Hermitian geometry and hence Hilbert-space quantum structure.

2. Quantum Reconstruction

[KNOWN/COND] With the relevant representation assumptions, physical propositions form the appropriate orthomodular/Sasaki structure and are represented by closed subspaces or projectors on a Hilbert space H .

[KNOWN/COND] Gleason-type representation gives probabilities:

$$\mu_P(P) = \text{Tr}(pP).$$

For a pure state $|\psi\rangle$ and rank-one projector $|\phi\rangle\langle\phi|$:

$$p(\phi|\psi) = |\langle\phi|\psi\rangle|^2.$$

[KNOWN/COND] Composition, local tomography, and the adopted compositional constraints select ordinary complex quantum composition:

$$HAB = HA \otimes_{\mathbb{C}} HB.$$

The real and quaternionic alternatives fail the required local-tomographic/compositional dimension matching under the assumptions used.

This is the first major structural result: quantum mechanics is not inserted as a starting ontology. It appears as the representational geometry of internally self-dual distinction.

3. Minimal Complex Distinction and Lorentz Geometry

Take the smallest nontrivial complex system:

$$H_2 = \mathbb{C}^2.$$

Its real vector space of Hermitian operators is:

$$V = \text{Herm}_2(\mathbb{C}).$$

[DER] Define the complement:

$$J(X) = \text{Tr}(X)I - X,$$

$$\text{with } J^2 = I.$$

Define:

$$g(X, Y) = \frac{1}{2} \text{Tr}[XJ(Y)].$$

Write:

$$X = x^0 I + x^i \sigma_i.$$

Then:

$$g(X, X) = (x^0)^2 - |\vec{x}|^2 = \det X.$$

Therefore:

$$\text{Herm}_2^*(\mathbb{C}) \cong L_4^+,$$

the four-dimensional future Lorentz cone.

[KNOWN] The action:

$$X \mapsto AXA^\dagger, \quad A \in \text{SL}(2, \mathbb{C}),$$

preserves $\det X$, and:

$$\text{SL}(2, \mathbb{C})/\{\pm I\} \cong \text{SO}^+(1, 3).$$

Thus the minimal complex binary distinction carrier has four real dimensions and possesses the local algebraic geometry of 3+1-dimensional Lorentzian spacetime.

Important scope statement: the qubit/Lorentz correspondence is known mathematics. The present contribution is its placement inside a distinction-first reconstruction, not a claim that the qubit/Lorentz correspondence itself is novel.

4. Primitive Distinction Dynamics

[DER] The unique covariant complementary qubit POVM under the adopted DSD “no new distinction from itself” condition is:

$$E(dQ) = 2(I-Q)d\nu_{FS}(Q).$$

For an existing pure distinction P , the induced outcome/source law is:

$$WP(dQ) = 2 \operatorname{Tr}[P(I-Q)] d\nu_{FS}(Q).$$

In Bloch-vector form:

$$w(n,m) = 1 - n \cdot m.$$

[DER] The associated Markov operator has:

$$\operatorname{spec}(T) = \{1, -1/3, 0\}.$$

The isotropic component is invariant; the $l=1$ orientation component decays in magnitude as 3^{-n} ; higher harmonics are killed in the idealized one-step source channel.

Critical correction:

3^{-d} is a multi-step distinction-information attenuation law. It is NOT a one-step probability for jumping record distance d .

The rejected one-step kernel $P_{xy} \propto 3^{-dR}(x,y)$ was shown to be nonlocal on a natural Boolean/Hamming record cube: one step changes an extensive fraction of the record coordinates.

5. Thermodynamic Realization and the Emergence of Events

Microscopic DSD evolution is reversible. Reversibility alone cannot select a unique causal arrow.

[DER/COND] A distinction becomes operationally realized when it is redundantly recoverable by many independent internal record fragments.

A model record isometry is:

$$VP(R) = P \otimes |1\rangle \otimes R + (I-P) \otimes |0\rangle \otimes R.$$

Only orthogonal/classical alternatives are redundantly broadcast, so this does not clone arbitrary unknown quantum states.

If fragment fidelity is bounded by $q < 1$, the failure of redundant distinguishability decays exponentially with the number of fragments:

$$FR \leq q^R.$$

In the thermodynamic limit, redundantly recorded alternatives can become disjoint sectors for finite local operations. This supplies operational persistence without violating microscopic reversibility.

[DEF] Let $\mathfrak{R}_e$ denote the objective record algebra of realized event e .

[DER] Record inclusion induces causal order:

$$e \leq f \Leftrightarrow \mathfrak{R}_e \subseteq \mathfrak{R}_f,$$

modulo equivalence of events with identical objective record content.

Refinement discovered later:

record incomparability alone is not sufficient for spacelike independence. True realized incomparability additionally requires no operational signaling and closure of the RPI diamond described below.

6. Causal Growth No-Go Results

The early growth model identified each microscopic binary DSD interaction with one realized causal event. That model fails.

Let k_n be the number of direct/maximal causal parents of realized event e_n . Under persistent growth, every parent-child relation is an empty interval/link and remains empty.

Therefore:

$$N_0(N) = N_0(\text{seed}) + \sum_n k_n.$$

If $k_n \leq K$ uniformly, then:

$$N_0(N) = O(N).$$

But the four-dimensional interval phase requires:

$$N_0(N) \sim C N^{3/2}.$$

Hence:

Bounded-Parent No-Go Theorem:

Any persistent realized-event growth law with a uniformly bounded number of direct parents cannot generate the four-dimensional causal-set interval phase.

The conceptual correction is fundamental:
binary DSD microinteraction $\neq$ thermodynamically realized causal event.

A realized event may summarize a larger pre-realization interaction complex and inherit an unbounded number of independent maximal histories.

7. Record Geometry and the Correct Primitive Locality

[DEF] For objective record sets:
 $dR(x,y) = |\mathfrak{R}x \Delta \mathfrak{R}y|$.

Primitive ancestry evolution must be local in record space.

[DER] The admissible microscopic ancestry graph is therefore the record cover graph:
 $x \sim y$ iff $dR(x,y)=1$,
or the corresponding weighted finite-range version.

A reversible local generator has form:
 $(\mathcal{L}f)(x) = \sum_{y \sim x} c_{xy} [f(y) - f(x)]$,
with $c_{xy} = c_{yx} \geq 0$.

This restores the correct logic:
local primitive step $\rightarrow$ many local steps $\rightarrow$ long-range correlation attenuation.

[KNOWN/COND] For reversible finite-range graph walks, volume doubling plus a scale-2 Poincaré inequality imply parabolic Harnack inequalities and Gaussian heat-kernel bounds. These conditions are not consequences of DSD alone; they must emerge from the realized vacuum geometry.

8. Operational Vacuum and Homogeneity

Do not interpret “vacuum” as “the universe must be empty.” The vacuum is the baseline no-excitation phase.

[DEF] Operational vacuum:
for every two contexts x,y and every finite observation radius R ,

$BR^*(x) \cong BR^*(y)$,

where the decorated rooted ball includes all locally observable record, DSD, and transition structure.

No finite internal experiment can determine where in the vacuum it is located.

[DER/KNOWN] In a connected locally finite setting, compatible rooted-ball equivalences at every finite radius extend to global automorphisms. Therefore operational vacuum implies homogeneity/vertex transitivity.

This removes homogeneity as an arbitrary geometric assumption: it follows from the operational meaning of “no localized vacuum observable.”

[KNOWN/COND] In the median/cubical record sector, homogeneous non-exponentially growing geometries reduce at large scale to lattice-like products $\mathbb{Z}^m$ times finite internal factors.

The dimension m remains unfixed at this step.

9. Why the Realized Vacuum Is Four-Dimensional

Assume the ancestry walk in the homogeneous lattice phase has the standard diffusive heat-kernel behavior:

$pt(x,x) \sim t^{-m/2}$.

Consider two independent ancestry histories. Their expected intersection count through scale L behaves as:

$E[IL] \sim$

$L^{2-m/2}$, $m < 4$;

$\log L$, $m = 4$;

$O(1)$, $m > 4$.

Thermodynamic realization requires two competing properties.

First, independent witnesses must become asymptotically independent at large separation. Below four dimensions, histories intersect too strongly for this.

Thus:

witness independence $\Rightarrow m \geq 4$.

Second, independent witnesses must remain indefinitely internally corroboratable as records of the same event. Above four dimensions, total intersection opportunity becomes finite.

Thus:

indefinite corroborability $\Rightarrow m \leq 4$.

Together:

$m = 4$.

At the critical point:

$E[|L|] \sim c_1 \log L \rightarrow \infty$,

while long-range intersection probability decays marginally, approximately as $c_2/\log L \rightarrow 0$.

So four is the unique critical phase where independent witnesses separate enough to be independently accessible but retain enough shared ancestry to remain indefinitely corroboratable.

This is independent of the earlier algebraic fact:

$\dim_{\mathbb{R}} \text{Herm}_2(\mathbb{C}) = 4$.

Therefore the theory has the cross-lock:

$d_{\text{algebra}} = d_{\text{realization}} = 4$.

10. Vacuum Stability

The isotropic primitive distinction distribution is invariant:

$TvFS = vFS$.

For zero-mean local orientation perturbations δf :

$\|T_n \delta f\|_2 \leq 3^{-n} \|\delta f\|_2$.

Therefore local primitive distinction-orientation perturbations are gapped/contractive rather than naturally gapless.

Label-covariant primitive evolution preserves exact vacuum symmetry if begun in the exact homogeneous vacuum sector.

Finite local modifications of a $\mathbb{Z}_4$ -like vacuum do not change its infrared quasi-isometry class or r_4 volume-growth exponent. Localized deviations can therefore be treated as excitations/defects rather than destruction of spacetime itself.

What is NOT claimed:

the four-dimensional vacuum has not been proven to be the unique global attractor of every arbitrary pre-spacetime initial condition. Vacuum stability is weaker and is the physically relevant requirement.

11. Causal-Set Continuum Diagnostics

For interval counts $N_m(C)$, define:

$$FC(z) = \sum_{m \geq 0} N_m(C) z^m.$$

Under independent Bernoulli thinning T_p :

$$E[FT_p C(z)|C] = p^2 FC(1-p+pz).$$

Assume scaling:

$$FN(z) \sim N_\gamma L(N) f(z).$$

Scale covariance gives:

$$p_\gamma f(z) = p^2 f(1-p+pz),$$

whose nontrivial solution is:

$$f(z) = C(1-z)^{-(2-\gamma)}.$$

Matching the known Minkowski interval family yields:

$$deff = 2/(2-\gamma).$$

For $d=4$:

$$\gamma = 3/2,$$

$$f^*(z) = C(1-z)^{-1/2},$$

and:

$$\begin{aligned} N_m/N_0 &\rightarrow \Gamma(m+1/2)/(\sqrt{\pi} \Gamma(m+1)) \\ &= (2^m \text{ choose } m)/4^m. \end{aligned}$$

The first ratios are:

$$1, 1/2, 3/8, 5/16, 35/128, \dots$$

This supplies a concrete continuum/manifold-likeness diagnostic rather than merely interpreting “four-dimensional” abstractly.

12. The Single Fundamental Dynamical Postulate: Realization-Path Independence

[POST] Realization-Path Independence (RPI):

For one completely realized intrinsic DSD-decorated causal history, different linear extensions or foliations used to compose mutually spacelike-independent realization moves must define the same physical map modulo internal gauge transformations that act trivially on fixed physical boundary observables.

RPI does NOT identify genuinely different microscopic histories, different spacetime paths, physical braids, or different curvature holonomies.

It identifies only different bookkeeping/slicing descriptions of the same intrinsic realized history.

Elementary diamond form:

for a stem l and realized independent events $e \parallel f$,

$$\begin{aligned} & [\mathcal{R}f(l+e) \circ \mathcal{R}e(l)]_{\text{phys}} \\ & = \\ & [\mathcal{R}e(l+f) \circ \mathcal{R}f(l)]_{\text{phys}}. \end{aligned}$$

True spacelike independence therefore requires:

no record ancestry $e \rightarrow f$,

no record ancestry $f \rightarrow e$,

no signaling $e \leadsto f$,

no signaling $f \leadsto e$,

and physical closure of the RPI diamond.

13. Quantum Coherence and the RPI Anomaly Test

Any two linear extensions of a finite partial order are connected by adjacent swaps of incomparable events. For an antichain, the swap algebra is generated by:

$$s_i^2 = 1,$$

$$s_i s_j = s_j s_i \text{ for } |i-j| > 1,$$

$$s_i s_{i+1} s_i = s_i s_{i+1} s_i s_{i+1} s_i.$$

Three-event projective $U(1)$ obstructions are removable.

For four or more events a nontrivial $\mathbb{Z}_2$ projective class exists abstractly for symmetric groups, so a central sign can appear at the vector-representation level.

However, the actual complex tensor composition of operationally independent realized sectors uses canonical factor swaps. These satisfy the symmetric-group relations exactly.

Even for a projective lift:

$$U_g U_h = \omega(g, h) U_{gh},$$

the induced physical channel is:

$$\text{Ad} U_g(\rho) = U_g \rho U_g^\dagger,$$

and:

$$\text{Ad} U_g \circ \text{Ad} U_h = \text{Ad} U_{gh},$$

because the central $U(1)$ phase cancels.

Therefore:

$$\text{HolRPI}_{\text{phys}} = 1$$

for the objective finite-dimensional DSD record sector.

A projective fermionic sign on state vectors need not be an observable foliation anomaly.

Physical exchange/braiding remains distinct from a bookkeeping linear-extension swap.

14. Earning the Hypersurface-Deformation Algebra

This is the key bridge from discrete RPI to continuum refoliation symmetry.

Take a smooth realized record front Σ_ℓ , a maximal antichain at coarse spacing ℓ .

Let its tangent DSD displacements be:

$$E_a = \ell e_a + O(\ell^2), \quad a=1,2,3.$$

The local Lorentz bilinear already derived from $\text{Herm}2(\mathbb{C})$ induces the spatial Gram metric:

$$q_{ab} = -g(e_a, e_b).$$

Let n be the unique future unit normal:

$$g(n, n) = 1,$$

$$g(n, e_a) = 0.$$

A local realization-front advance with arbitrary lapse $N(x)$ is:

$$\delta N X = N n.$$

Preserving normalization implies $\delta N n$ is tangent. Write:

$$\delta N n = A_a e_a.$$

Preserving orthogonality:

$$0 = \delta N g(n, e_a)$$

$$= g(\delta N n, ea) + g(n, \delta N ea).$$

Since:

$$\begin{aligned}\delta N ea &= \nabla a(Nn) \\ &= (\partial a N)n + N \nabla a n,\end{aligned}$$

we obtain:

$$g(n, \delta N ea) = \partial a N.$$

Also:

$$g(\delta N n, ea) = -q_{ab} A^b.$$

Therefore:

$$A^b = q^b_a \partial a N,$$

or:

$$\delta N n = q_{ab} \partial^b N ea,$$

where q_{ab} denotes the inverse spatial metric.

Now compare two realization orders:

$$\delta N \delta M X = M \delta N n,$$

$$\delta M \delta N X = N \delta M n.$$

Their commutator is:

$$[\delta N, \delta M]X$$

=

$$q_{ab}(M \partial^b N - N \partial^b M) ea.$$

Define the tangential shift:

$$V_a = q_{ab}(M \partial^b N - N \partial^b M).$$

Thus:

$$[\delta N, \delta M] = \delta V.$$

Together with ordinary tangential relabelings:

$$[\delta \vec{N}, \delta \vec{M}] = \delta[\vec{N}, \vec{M}],$$

$$[\delta \vec{N}, \delta M] = \delta \mathcal{L}_{\vec{N}} M,$$

we obtain the complete hypersurface-deformation algebra geometrically.

The crucial structure function q_{ab} was not inserted from ADM. It arose because the DSD-induced spatial metric is the unique map converting the lapse gradient covector into the tangent-vector tilt of the future normal.

Discrete finite-difference form:

$$\begin{aligned} & \Delta X v \\ & = \\ & \varepsilon^2 q_{ab} [M_v \Delta b \ell N_v - N_v \Delta b \ell M_v] e_{a,v} \\ & + O(\varepsilon^2 \ell, \varepsilon^3). \end{aligned}$$

Hence the continuum bracket is recovered as $\ell/L \rightarrow 0$ in the smooth scaling sector.

15. Canonical Semiclassical Sector and HKT

[COND] The smooth geometric sector must admit a regular semiclassical symplectic description after gauge quotient.

Complex projective Hilbert space naturally carries a Kähler/symplectic structure. On a regular coherent geometric sector, local Darboux coordinates may be chosen as:
(q_{ab} , π_{ab}).

Let $D[\vec{N}]$ and $H[N]$ generate tangential and normal record-front deformations through Poisson brackets.

Because the generator brackets reproduce the geometric deformation commutators, the physical constraints satisfy:

$$\{D[\vec{N}], D[\vec{M}]\} = D[[\vec{N}, \vec{M}]],$$

$$\{D[\vec{N}], H[M]\} = H[\mathcal{L}_{\vec{N}} M],$$

$$\{H[M], H[N]\}$$

=

$$D[q_{ab}(M \partial_b N - N \partial_b M)].$$

[KNOWN/COND] Under the Hojman–Kuchař–Teitelboim locality/geometrodynamical assumptions for a metric-only canonical sector, representing this hypersurface-deformation algebra yields the ADM gravitational constraints.

Momentum constraint:

$$\mathcal{H}_a = -2 q_{ab} \nabla_c \pi^{bc}.$$

Hamiltonian constraint:

$$\mathcal{H}_\perp$$

=

$$\begin{aligned} & 16\pi G / \sqrt{q} (\pi_{ab} \pi^{ab} - 1/2 \pi^2) \\ & - \sqrt{q} / (16\pi G) ({}^{(3)}R - 2\Lambda). \end{aligned}$$

Canonical action:

SADM

=

$\int dt d^3x$

$[\pi_{ab} \dot{q}_{ab} - N \mathcal{H} - N^a \mathcal{H}_a]$.

This is equivalent to the Einstein–Hilbert action:

SEH

=

$1/(16\pi G)$

$\int d^4x \sqrt{-g} (R - 2\Lambda),$

and therefore, including matter:

$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$

at leading infrared order.

Higher microscopic corrections are expected beyond the continuum limit.

16. What Has Been Structurally Earned

The present structural chain is:

distinction

→ DSD adjoint structure

→ quantum/Hilbert representation under stated reconstruction conditions

→ complex composition

→ minimal qubit

→ $\text{Herm2}(\mathbb{C})$

→ local Lorentz cone

→ primitive complementary distinction dynamics

→ redundant thermodynamic records

→ persistent realized events

→ causal partial order

→ local record-cover ancestry dynamics

→ operational homogeneous vacuum

→ four-dimensional critical realized vacuum

→ causal-set 4D continuum diagnostics

→ RPI

→ anomaly-free physical swap coherence

→ DSD-induced record-front metric

→ hypersurface-deformation algebra

- HKT/ADM under canonical continuum conditions
- Einstein gravity in the infrared.

The single explicit fundamental dynamical postulate is RPI.

The dimension four is not inserted into RPI, the qubit carrier, or the random-walk intersection argument. It appears independently in the local complex-binary algebra and in the thermodynamic realization criticality.

17. No-Go Results and Removed Branches

The following branches have been explicitly rejected and must not silently re-enter later calculations.

1. Bounded realized-event parent number:
 $kn \leq K \Rightarrow N_0(N)=O(N)$, so it cannot produce the required four-dimensional interval phase.
2. Identifying one binary microscopic DSD interaction with one realized causal-set event:
 rejected. Microinteraction and thermodynamic realization occur on different levels.
3. Using 3-dR as a one-step ancestry jump kernel:
 rejected as nonlocal on natural record spaces. 3-d is multi-step information attenuation.
4. Assuming ordinary mixing automatically prevents history condensation:
 false. Rapidly mixing chains can still possess highly nonuniform stationary measures unless reciprocity/normalization structure is specified.
5. Requiring global ancestry equilibrium before realization:
 unnecessary and not generally guaranteed.
6. Defining spacelike independence solely as record incomparability:
 insufficient. Noncommuting local quantum updates provide an explicit counterexample.
7. Treating RPI as literal commutativity of all realization moves:
 wrong. Normal deformations close into tangential gauge, and genuinely different physical histories remain distinguishable.
8. Treating curvature path dependence as an RPI violation:
 wrong. Physical spacetime paths and bookkeeping/foliation paths are different objects.
9. Assuming discrete RPI immediately equals ADM refoliation symmetry:

too fast. The metric-dependent bracket had to be derived from DSD record-front normal geometry.

10. Claiming DSD kinematics alone specifies a microscopic geometric Hamiltonian: false. RPI supplies the irreducible dynamical principle, while the continuum deformation algebra plus HKT constrains the infrared gravitational dynamics.

18. Current Open Quantitative Frontier

The structural theory is frozen at this point. The next project is microscopic calibration.

[OPEN] Newton's constant G .

The framework presently fixes the form of the infrared gravitational kinetic term but not its absolute coefficient. G should be related to a microscopic stiffness, susceptibility, or response coefficient of the realized record vacuum.

[OPEN] Cosmological constant Λ .

The framework permits a vacuum curvature term but does not yet derive its value or sign. Λ should be related to the vacuum record free-energy density, constant-curvature holonomy density, or another intrinsic vacuum quantity.

[OPEN] Microscopic scale ℓ^* .

The record/coarse-graining scale that controls the onset of continuum behavior has not yet been calibrated to a physical length or energy.

[OPEN] Finite- ℓ corrections.

The discrete normal-deformation calculation contains corrections schematically of order ℓ/L , and the effective gravitational action contains higher-derivative terms schematically of order E^2/M^{*2} . Their coefficients must be derived from the microscopic theory rather than guessed.

[OPEN] Quantitative matter sector.

Localized record-symmetry defects provide a structural location for matter/excitations, but particle content, masses, gauge groups, couplings, and the Standard Model are not yet derived here.

[OPEN] Full global cosmological selection.

A stable four-dimensional vacuum phase has been identified structurally, but the theory has not proved that every arbitrary pre-spacetime initial state dynamically selects that phase.

19. Master Structural Equation

The current theory can be compressed as:

$(X, \perp)$

$\Rightarrow \text{DSD}$

$[f(a) \perp b \Leftrightarrow a \perp f(b)]$

$\Rightarrow$ quantum distinction geometry

$\Rightarrow H\mathbb{C}$

$\Rightarrow H^2 = \mathbb{C}^2$

$\Rightarrow \text{Herm}_2(\mathbb{C})$

$\Rightarrow L_{4+}$

$\Rightarrow$ local Lorentz structure

plus

$E(dQ) = 2(I - Q)d\nu_{FS}(Q),$

$WP(dQ) = 2\text{Tr}[P(I - Q)]d\nu_{FS}(Q),$

$\text{spec}(T) = \{1, -1/3, 0\}$

$\Rightarrow$ local distinction propagation

$\Rightarrow$ redundant thermodynamic recording

$\Rightarrow$ realized record sectors

$\Rightarrow$ causal partial order

$\Rightarrow$ local record-cover ancestry graph

$\Rightarrow$ operational vacuum homogeneity

$\Rightarrow$ realization criticality

$\Rightarrow d\nu_{ac} = 4$

$\Rightarrow$ smooth record fronts with

$q_{ab} = -g(e_a, e_b),$

$g(n, n) = 1,$

$g(n, e_a) = 0$

plus RPI

$\Rightarrow$

$\delta N_n = q_{ab} \partial_b N_{ea}$

$\Rightarrow$

$[\delta N, \delta M]$

$=$

$$\delta V,$$

$$V_a = q_{ab}(M \partial_b N - N \partial_b M)$$

together with tangential brackets

$\Rightarrow$ hypersurface-deformation algebra

$\Rightarrow$ HKT

ADM constraints

$\Rightarrow$

$$S_{EH} = (16\pi G)^{-1} \int \sqrt{-g} (R - 2\Lambda) d^4x$$

$\Rightarrow$

$$G_{\mu\nu} + \Lambda g_{\mu\nu}$$

=

$$8\pi G T_{\mu\nu}$$

+

microscopic corrections.

20. Current Claim Boundary

The framework should presently be described as a candidate structural reconstruction, not as a completed fundamental theory and not as a theory of everything.

The strongest defensible claim is:

Starting from self-dual internal distinguishability and one explicit dynamical principle—Realization-Path Independence—the framework provides a coherent conditional reconstruction chain connecting quantum distinction geometry, local Lorentz structure, thermodynamic causal realization, a four-dimensional homogeneous vacuum phase, refoliation symmetry, and Einstein gravity in the smooth metric-only infrared sector.

The quantitative values of G , Λ , the microscopic scale, correction coefficients, and the detailed matter sector remain open.

That boundary is intentional. Future calculations should either derive these quantities from the frozen structural theory or falsify one of its stated conditions; they should not repair failures by silently inserting the desired continuum answer.